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```
import numpy as np
import pandas as pd

# Input data files are available in the read-only "../input/" directory
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

Importing Necessary Libraries

```
import matplotlib.pyplot as plt
import cv2
import numpy as np
from keras.models import Sequential
from keras.layers import Dense, Flatten, Conv2D, MaxPool2D, Dropout
from keras.optimizers import SGD, Adam
from keras.callbacks import ReduceLROnPlateau, EarlyStopping
from keras.utils import to_categorical
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.utils import shuffle
```

```
!pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (2.0.0)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle) (6.3.0)
Requirement already satisfied: kagglesdk<1.0,>=0.1.15 in /usr/local/lib/python3.12/dist-packages (from kaggle) (0.1.16)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kaggle) (26.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle) (5.29.6)
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kaggle) (4.67.3)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.5.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from bleach->kaggle) (0.5.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil->kaggle) (1.17.0)
Requirement already satisfied: text-unidecode>=1.3 in /usr/local/lib/python3.12/dist-packages (from python-slugify->kaggle
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle) (3.11)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle) (2026
```

Read The Dataset

```
# Download the Kaggle dataset
!kaggle datasets download -d sachinpatel21/az-handwritten-alphabets-in-csv-format
```

```
Dataset URL: https://www.kaggle.com/datasets/sachinpatel21/az-handwritten-alphabets-in-csv-format
License(s): CC0-1.0
Downloading az-handwritten-alphabets-in-csv-format.zip to /content
100% 185M/185M [00:02<00:00, 71.8MB/s]
```

```
# Unzip the dataset files
!unzip az-handwritten-alphabets-in-csv-format.zip

# Read the CSV file into a pandas DataFrame
data = pd.read_csv("A_Z Handwritten Data.csv").astype('float32')
```

```
Archive:  az-handwritten-alphabets-in-csv-format.zip
  inflating: A_Z Handwritten Data.csv
  inflating: A_Z Handwritten Data/A_Z Handwritten Data.csv
```

```
data.head()
```

	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	...	0.639	0.640	0.641	0.642	0.643	0.644	0.645	0.646	0.647	0.64
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

5 rows × 785 columns

```
data.tail()
```

	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	...	0.639	0.640	0.641	0.642	0.643	0.644	0.645	0.646	0.647
372445	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
372446	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
372447	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
372448	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
372449	25.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

5 rows × 785 columns

Splitting Dataset into Labels and Images

```
X = data.drop('0',axis = 1) # axis=1 for dropping column
y = data['0']
```

```
X.head()
```

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.10	...	0.639	0.640	0.641	0.642	0.643	0.644	0.645	0.646	0.647	0.6
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0

5 rows × 784 columns

It is a 28x28 image flattened to 784 pixels

```
y.head()
```

	0
0	0.0
1	0.0
2	0.0
3	0.0
4	0.0

dtype: float32

Reshaping the data to display it as image and splitting the Dataset as Test and Train data

```
X_train, X_test, Y_train, Y_test = train_test_split(X, y, test_size = 0.2)

X_train = np.reshape(X_train.values, (X_train.shape[0], 28,28))
X_test = np.reshape(X_test.values, (X_test.shape[0], 28,28)) #0=> B&W, 28x28 matrix

```

```
print("Train data shape: ", X_test.shape)
print("Test data shape: ", X_test.shape)
```

```
Train data shape:  (74490, 28, 28)
Test data shape:  (74490, 28, 28)
```

Creating dictionary of column number to corresponding letter

```
word_dict = {0:'A',1:'B',2:'C',3:'D',4:'E',5:'F',6:'G',7:'H',8:'I',9:'J',10:'K',11:'L',12:'M',13:'N',14:'O',15:'P',16
```

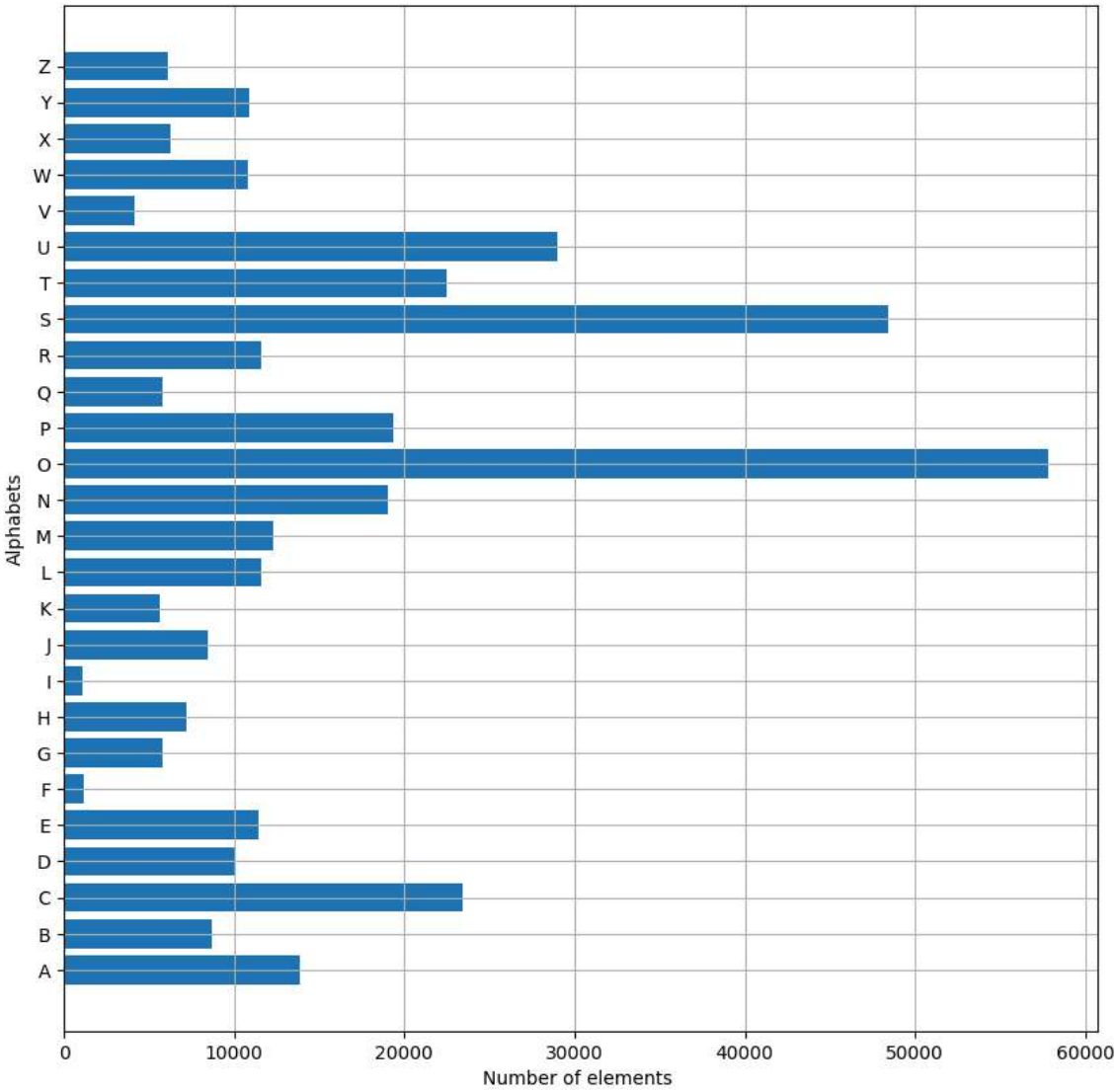
Plotting Count of Each Alphabet in Dataset

```
y_int = y.astype(int)
count = np.zeros(26, dtype='int') #a vector of size 26 with all 0 values
for i in y_int:
    count[i] +=1 #total count of each alphabet

alphabets = []
for i in word_dict.values():
    alphabets.append(i) #all alphabets

fig, ax = plt.subplots(1,1, figsize=(10,10))
ax.barh(alphabets, count)

plt.xlabel("Number of elements ")
plt.ylabel("Alphabets")
plt.grid()
plt.show()
```

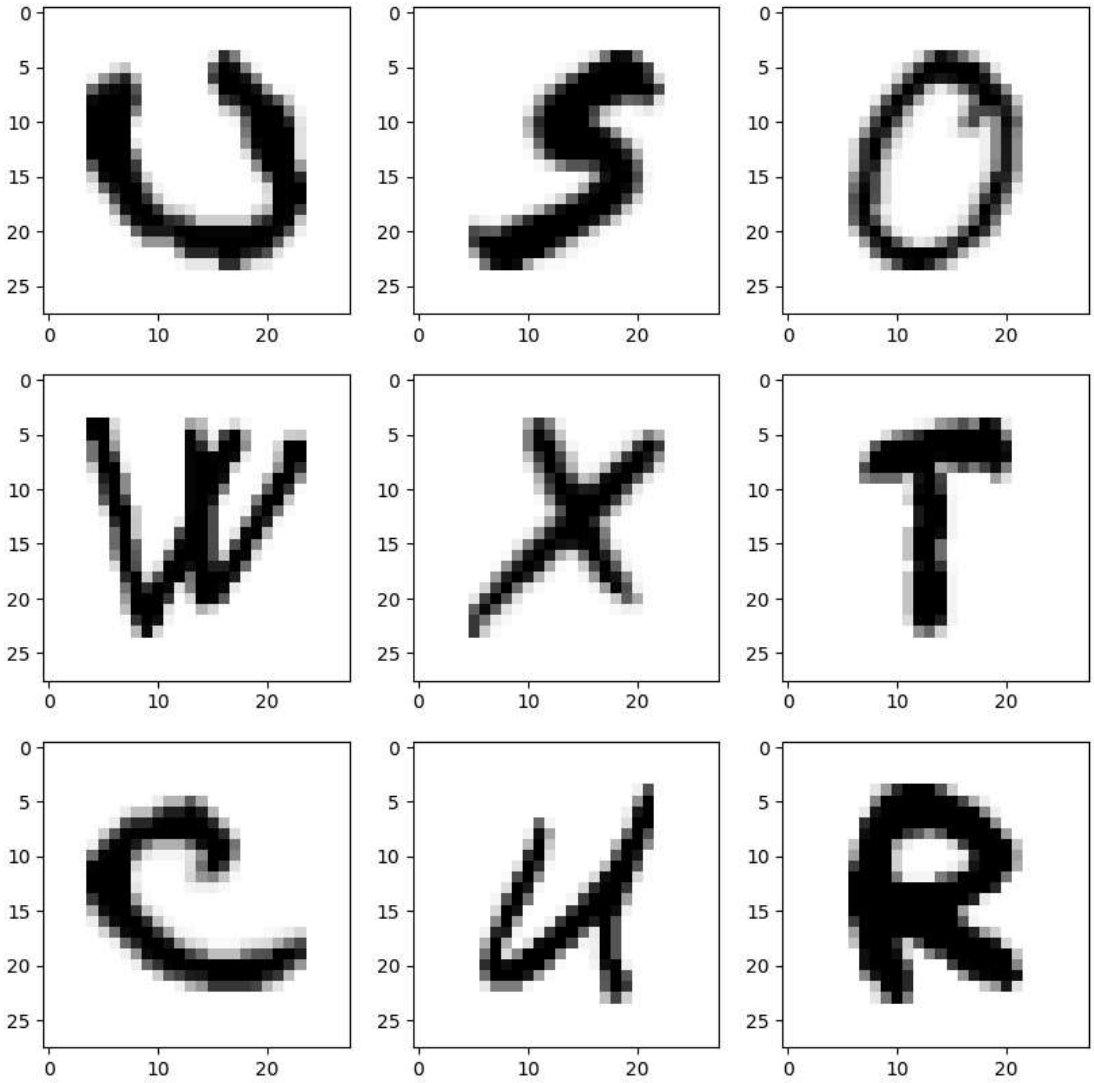


Displaying few letters

```
shuff = shuffle(X_train[:100])

fig, ax = plt.subplots(3,3, figsize = (10,10))
axes = ax.flatten()

for i in range(9):
    _, shu = cv2.threshold(shuff[i], 30, 200, cv2.THRESH_BINARY)
    axes[i].imshow(np.reshape(shuff[i], (28,28)), cmap="Greys")
plt.show()
```



Reshaping data to feed to CNN

```
X_train = X_train.reshape(X_train.shape[0],X_train.shape[1],X_train.shape[2],1) #RGB =>Channel of 1
print("New shape of train data: ", X_train.shape)

X_test = X_test.reshape(X_test.shape[0], X_test.shape[1], X_test.shape[2],1) #RGB =>Channel of 1
print("New shape of train data: ", X_test.shape)
```

```
New shape of train data:  (297960, 28, 28, 1)
New shape of train data:  (74490, 28, 28, 1)
```

Converting Single float value to Categorical Values

```
train_yOHE = to_categorical(Y_train, num_classes = 26)
print("New shape of train labels: ", train_yOHE.shape)

test_yOHE = to_categorical(Y_test, num_classes = 26)
print("New shape of test labels: ", test_yOHE.shape)
```

```
New shape of train labels:  (297960, 26)
New shape of test labels:  (74490, 26)
```

Building CNN Model

```
model = Sequential()
#CNN
# input -> conv -> maxpool -> conv -> maxpool .....->flattened vector->
#.                        hidden layer -> hidden layer -> softmax layer
model.add(Conv2D(filters=32, kernel_size=(3, 3), activation='relu', input_shape=(28,28,1)))
model.add(MaxPool2D(pool_size=(2, 2), strides=2))

model.add(Conv2D(filters=64, kernel_size=(3, 3), activation='relu', padding = 'same'))
model.add(MaxPool2D(pool_size=(2, 2), strides=2))

model.add(Conv2D(filters=128, kernel_size=(3, 3), activation='relu', padding = 'valid'))
model.add(MaxPool2D(pool_size=(2, 2), strides=2))

model.add(Flatten())

model.add(Dense(64,activation ="relu"))
model.add(Dense(128,activation ="relu"))

model.add(Dense(26,activation ="softmax"))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `inpu
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Compiling and Training the CNN

```
model.compile(optimizer = Adam(learning_rate=0.001), loss='categorical_crossentropy', metrics=['accuracy'])

history = model.fit(X_train, train_yOHE, epochs=1,  validation_data = (X_test,test_yOHE))
```

```
9312/9312 ━━━━━━━━━━━━━━━━━━━━ 387s 41ms/step - accuracy: 0.9568 - loss: 0.1590 - val_accuracy: 0.9765 - val_loss: 0.0870
```

Printing Model Summary

```
model.summary()
model.save(r'model_hand.h5')
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 2, 2, 128)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 64)	32,832
dense_1 (Dense)	(None, 128)	8,320
dense_2 (Dense)	(None, 26)	3,354

Total params: 411,536 (1.57 MB)
Trainable params: 137,178 (535.85 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 274,358 (1.05 MB)

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file f
```

```
print("The validation accuracy is :", history.history['val_accuracy'])
print("The training accuracy is :", history.history['accuracy'])
print("The validation loss is :", history.history['val_loss'])
print("The training loss is :", history.history['loss'])
```

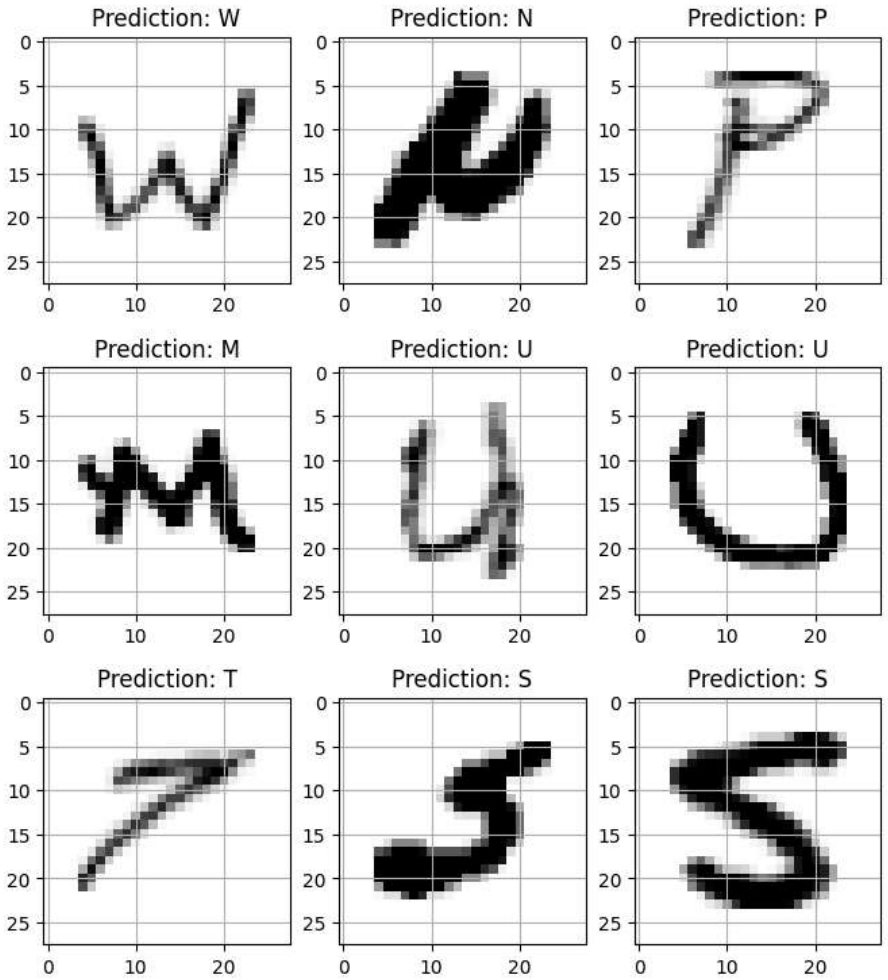
```
The validation accuracy is : [0.9765471816062927]
The training accuracy is : [0.9568465352058411]
The validation loss is : [0.0869632288813591]
The training loss is : [0.15895158052444458]
```

Making Predictions

```
fig, axes = plt.subplots(3,3, figsize=(8,9))
axes = axes.flatten()

for i,ax in enumerate(axes):
    img = np.reshape(X_test[i], (28,28))
    ax.imshow(img, cmap="Greys")

    pred = word_dict[np.argmax(test_yOHE[i])]
    ax.set_title("Prediction: "+pred)
    ax.grid()
```




```
Start coding or generate with AI.
```